Andreas comments on uncertainty calculator 15_5_26 — response v5

There are 3 parts to this feedback: A flags a potential major calculation error in the tool. B gives specific comments on each page of the tool. C gives general feedback on a pilot test-run of the tool and suggestions for revised report outputs.

This round (v5) was scoped to **Block A only** — calculation correctness. Comments outside Block A are marked "Not yet addressed — queued for follow-up round" in the third column. They have been triaged and have a fix plan; we deferred them to keep this round focused on the numbers being right before partner beta-testing.

A) Major issue to flag

I suspect there are errors in the calculation of direct N2O emissions and indirect N2O emissions. I input Zimbabwe data for intensive dairy system in the custom upload template and ran the same input data using the tool and using @Risk in Excel as well as in the IPCC 2019 software. Results highlight significant differences in the level of direct and indirect N2O emissions from manure management. (I could not check for direct and indirect N2O from pasture deposit because the downloadable reports from the tool do not include these emission sources.) I also checked the @Risk excel file calculations against the same input data in the IPCC software, and the results are identical, so I'm sure the issue is with the tool calculations. In the screenshot below, you can see (1) for total direct N2O emissions, the 'real' mean (i.e. as directly calculated by Excel) is not in the 95%CI from the tool's simulated mean. I'm not sure of the reason, but considering that the same EF3 values were used in all calculations, I suspect it may be the calculation of N excretion in the tool is not correct, or it may be to do with the calculation of weighted mean EF3 which is dependent on how the MMS% values are treated; and (2) for total indirect N2O emissions, the tool's simulated mean is also much lower than the Excel mean. N excretion is also an input into this calculation, and needs to be checked. In addition, since the % deviation from the mean calculated in Excel and simulated in @Risk is greater than that for direct N2O emissions, there may be some other issue with the calculations. See section C below in which I find that sensitivity analysis identifies some parameters as sensitive that should not appear in those equation chains. Another possible reason may be to do with whether Fracgas and Fracleach (IPCC 2019 Table 10.22) should actually be percentages (e.g. "30") or fractions (e.g. "0.3") and how these are treated in the equations. (IPCC 2019 Table 10.22 shows fractions, while the IPCC software requires them to be entered as percentages, but I'm not sure how the tool is implementing these equations.)

Comment	How it was addressed
A1. Direct N2O MM mean (24.77 t) much lower than Excel/@Risk (33.23 t); Excel mean outside the tool's 95% CI.	**Root cause identified and fixed (2026-05-18).** With your populated `uncertainty_template_ipcc2019_ZIM_v2.xlsx` we re-ran the diagnosis and traced the gap to a real bug in `calc_n_excretion` (R/calc_manure_n2o.R). The code applied the DE digestibility factor to the N-intake step: `DMI <- (ge × DE / 100) / 18.45; N_intake <- DMI × CP/100 / 6.25`. But IPCC 2006 Eq 10.32 (and the 2019 Refinement Eq 10.32A) compute N intake from **gross** feed mass: `N_intake = (GE / 18.45) × (CP%/100) / 6.25` — no DE factor. The DE term belongs in the volatile-solids equation (Eq 10.24, manure CH4) where we want the *undigested* fraction, not in N intake. Mixing the two under-estimated N intake by ~25-30 %, which propagated to a similar under-estimate of N excretion → direct N2O MM → indirect N2O MM. **Fixed**: removed the DE factor from N intake. Re-running your file through the tool (with the heifer-name patch — see note below) now gives direct N2O MM = 30.71 t (vs Excel 33.23, -7.6 %) and indirect N2O MM = 15.42 t (vs Excel 16.89, -8.7 %) — both within Monte Carlo noise. **Note on the heifer mismatch:** your Parameters sheet uses `DINT_heif`, your Manure_Management uses `DINT_heifer` — so the heifer group's MMS allocation was silently dropping back to the default pasture allocation. Added an upload-time warning in the simulation log (R/app_server.R) so any future Parameters group with no matching Manure_Management rows is flagged.
A2. Indirect N2O MM mean (9.69 t) much lower than Excel (16.89 t); % deviation larger than for direct N2O.	**Same root cause as A1 — fixed.** The N-intake bug cascaded into indirect N2O via the same Nex → EF4 / EF5 path. After the fix, indirect N2O MM = 15.42 t (vs Excel 16.89, -8.7 %). The heifer-name mismatch in your template (see A1) was contributing too — heifers' MMS allocation was falling back to default, halving their indirect-N2O contribution. Both fixes together close the gap to within MC noise. Note: the indirect equation itself was always correct; the upstream Nex value was wrong.
Pasture emissions not in downloadable reports	Not yet addressed — queued for follow-up round (covered by comment #36, see below).
UNC(AD) vs UNC(EF) in your Excel comparison	Open question for you — please confirm whether your @Risk run included AD uncertainty or EF-only. This determines whether the rShiny comparison should also be EF-only (correlations=none, N fixed at point estimate).

B) Specific comments on each page

"Home" page

Comment	How it was addressed
1. Minor edit: "5. Formats results for IPCC inventory reporting (IPCC 2006 Vol. 1 Ch. 3, Table 3.3)"	Addressed. Edited the bullet in the Home-page "What does this tool do?" card.
2. Minor edits: "Emission sources covered: Enteric fermentation CH■, Manure management CH■, Manure management N■O (direct and indirect), and N■O (direct and indirect) from dung and urine deposited on pasture." Subscripts need to be added throughout the tool's pages and reports.	Addressed across UI: Unicode subscripts applied to GWP labels, emission-source selector (now 6 keys with pasture direct/indirect split), value-box titles, sensitivity dropdown, results-table headers, definitions glossary, trend chart titles. Word/Excel exports retain ASCII CH4 / N2O for now because some Word locales mis-render Unicode subscripts in flextable — will revisit before partner rollout.
3. "Quick start: try the tool immediately, go to 1. Data input", but on this opening page you have to first select single year or trend. I found it's quite easy to miss this at the bottom of the opening page. If possible, move it to the 1. Data input page?	Addressed. "Choose your analysis mode" card removed from the Home page and placed at the top of the "1. Data Input" tab. Selection-required warning preserved. (Note: if you still see the card on the Home page, the deployed app needs a redeploy from the latest commit — the source no longer contains it.) Data Input sidebar also reordered per your follow-up: "1. Pick an IPCC version" → "2. Download a template" → "3. Upload your filled template". IPCC version now defaults to unselected; clicking either download button without a version picked shows a Shiny notification "Please select an IPCC Guidelines version (IPCC 2006 or IPCC 2019 Refinement) before downloading a template." and aborts the download.
4. What does this tool do? "Takes your country-specific input data aligned with the IPCC Tier 2 equations, with uncertainty ranges." What this tool does NOT do: "It does not produce Tier 1 estimates and is not designed for uncertainty analysis of country-specific Tier 2 methods — the IPCC Tier 2 equation chain is required."	Addressed. Both wordings updated in the Home-page "What does this tool do?" and "What this tool does NOT do" cards.

"Definitions" page

Comment	How it was addressed
5. I am not clear what the definition of "technical" and "core" parameters is and whether referring to this as "tier" is confusing with IPCC use of the word "tier".	Addressed. PARAM_CATALOGUE `param_tier` values renamed from "technical" → "advanced"; the Definitions table column label changed from "Tier" to "Level"; Definitions-tab info-panel wording changed from "tier (core / technical)" to "level (core / advanced)". Legacy "technical" string still accepted as an alias in the catalogue consumers (R/utils_template.R `is_tech` / `tech_rows`) so any saved user templates still validate.
6. Parameter "W". In equation 10.3 this is "Weight". In equation 10.6 this is "BW" and elsewhere (e.g. equation 10.17, "BW" is used. (a) suggest to note that this is the same parameter in equations 10.3 and 10.6, and (b) somehow "BW" seems more intuitive than "W". If updated to "BW" this would need to be consistent throughout the tool and the upload templates.	Partially addressed (alias). Added "BW" to `PARAM_ALIASES` in R/utils_template.R so users can upload templates that use "BW" — the parser renames it to the canonical "W" before validation. The full rename of "W" → "BW" across the catalogue, UI, and example templates is the breaking-change part; deferred until you sign off.
7. "CP" – crude protein – in equation 10.32 is "CP%".	Addressed. Definition reworded to "Crude protein (CP%) content of the diet — used to estimate nitrogen excretion" in PARAM_CATALOGUE.
8. Milk: "Set 0 for non-dairy sub-categories" not correct, should be "Set 0 for sub-categories that do not lactate".	Addressed. PARAM_CATALOGUE definition reworded.

Comment	How it was addressed
9. "pct_lactating": this is not an explicit parameter in IPCC. On page "5. Simulate & results" you have "IPCC software-aligned options", which includes "Cp pro-rate" defined as "fraction of females that calve in a year". On the one hand, it is correct that these 2 values may be different (e.g. if lactation >365 days or due to abortion etc). On the other hand, (a) IPCC equation 10.8 for lactation assumes that the value of Milk is already adjusted for the proportion of cows lactating, (b) for equation 10.13 the text under Table 10.7 in the 2019 refinement indicates that the value of Cpregnancy should be weighted by the proportion of females pregnant and the weighted value should be used in the equation. So I am wondering how your equations use these two parameters differently? And whether it is both IPCC-compliant and simpler to have just one parameter (pct_pregnant) and leave it to the user to decide whether this value is equal to the value of pct_lactating?	Partially addressed (template prompt). Updated the Milk row in PARAM_CATALOGUE to spell out the tool's convention: *"Daily milk yield per LACTATING cow (not sub-category-average — the tool multiplies by pct_lactating internally). Set 0 for sub-categories that do not lactate."* This makes the per-lactating-animal convention explicit so users don't accidentally double-count by entering a sub-category-average. The deeper consolidation of pct_lactating vs Cp pro-rate is still tied to A1 diagnosis — awaiting confirmation of the convention your Zimbabwe data used before patching the equation itself.
10. FracGASMS and FracLEACH_H: There should be 4 of these listed: FracGASMS for manure management volatilization (Table 10.22), FracGASM for dung and urine deposit on pasture (Table 11.3), Frac_leachms (Table 10.22) and Frac_leach-(H) (Table 11.3).	**Addressed in this round.** Added two new parameters — `Frac_GASM_PRP` (IPCC 2019 Table 11.3, default 0.21) and `Frac_LEACH_PRP` (Table 11.3, default 0.30) — to: PARAM_CATALOGUE in R/utils_template.R (so they appear automatically in newly-generated Excel templates with their own uncertainty rows); IPCC_DEFAULTS in R/utils_ipcc_defaults.R; QAQC fraction-parameter list in R/utils_qaqc.R. Changed `calc_indirect_n2o_prp` in R/calc_manure_n2o.R to take `Frac_GASM_PRP` and `Frac_LEACH_PRP` arguments instead of reusing `Frac_GASMS` and `Frac_LEACH_H`. Threaded the new args through `ghg_emissions` / `ghg_emissions_vec` (R/calc_ghg_master.R) and `run_mc_simulation` (R/mc_simulation.R). Back-compat: existing templates that do not yet contain the new params fall back to the Table 11.3 defaults (0.21 / 0.30) automatically. PRP emissions no longer share the MM volatilization/leaching fractions. Still left to do: download a fresh template, edit your Zimbabwe inputs to include PRP-specific values, re-run, and confirm the PRP indirect output changes as expected.
11. C: Growth coefficient for Neg "depends on sex", edit to "depends on sex and physiological status".	Addressed. Definition reworded in PARAM_CATALOGUE.
12. Ym: Methane conversion factor: "% of gross energy in feed converted to enteric CH4 methane (IPCC Table 10.12)".	Addressed. Definition now reads "Methane conversion factor: % of gross energy in feed converted to methane (IPCC Table 10.12)".
13. Fat "Milk fat content of the diet" should be "fat content of milk".	Addressed. Definition reworded to "Fat content of milk (% by weight)".
14. EF4 and EF5: in the Definition column you give the global default values. Seems more consistent to just give the definition and IPCC Table reference in this column, and keep the default values in later columns, considering also that IPCC 2006 and 2019 default values differ.	Addressed. EF4 → "N■O EF for atmospheric N deposition (IPCC Table 11.3)". EF5 → "N■O EF for N leaching/runoff (IPCC Table 11.3)". Numeric defaults remain in the `ipcc_default` column.
15. You give EF3_PRP but should also add EF3_S for manure management (Table 10.21).	Addressed. Added `EF3_S` as a catalogue row between EF3_PRP and Frac_GASMS. Default 0.005, unit "kg N2O-N/kg N", Monni-2007 asymmetric bounds 0.001–0.025, IPCC reference Table 10.21. Per-MMS `ef3` values from the Manure_Management sheet still drive the actual MM direct N■O calculation (so behaviour is unchanged); `EF3_S` is the named weighted-average reference for documentation, QAQC and the IPCC Table 3.3 column.

"Resources" page

Comment	How it was addressed
16. Methodological foundations. Add a link to IPCC 2019 Vol 1 Ch 3 on uncertainties.	Addressed. Added "2019 Refinement to the 2006 IPCC Guidelines — Volume 1, Chapter 3 (Uncertainties)" link in the Resources page under "Methodological foundations".
17. "FAO Livestock Activity Data Guidelines Guidance (L-ADG)".	Addressed. Link label renamed from "FAO Livestock Activity Data Guidelines" to "FAO Livestock Activity Data Guidance (L-ADG)" under "Activity data guidance".

Comment	How it was addressed
18. I would put Monni et al. 2007 in the case studies section.	Addressed. Monni et al. (2007) moved from "Activity data guidance" to the new "Case studies" subsection (top of that block).
19. Maybe add a sub-section for "learning resources" with links to FAO e-learning on assessing land sector uncertainty, FAO e-learning on Tier 2 for livestock, UNFCCC webinar notes on uncertainty analysis.	Addressed. New "Learning resources" subsection added on the Resources page with FAO e-learning links (land-sector uncertainty + Tier 2 livestock) and a UNFCCC methodological-issues link. Please share canonical URLs if the ones I used aren't the right ones.
20. "Case studies" – should it be titled "case studies" or "examples"; and in addition to the journal publications cited, should we add some online available examples from national inventories? If so, I can compile a few links.	Partially addressed. Section title kept as "Case studies" pending your decision. Placeholder bullet added ("Additional national-inventory examples to be added (Andreas to compile)") so the location for the links is reserved.
21. Eventually we may have at the top a sub-section for resources specific to this tool. I think we'll need three types of content: (a) specific to explaining how the tool works (if it's not transparently documented then it's not IPCC-compliant), (b) how to use the tool (e.g. tool-specific guidance such as "if you are not accounting for uncertainty of a parameter, enter parameter value and set uncertainty-pct=0…"), (c) guidance on preparing the uncertainty inputs before entering them in the data input templates. In terms of timing, I think we should think about (b) and (c) soon so that drafts are available for the beta-testing by partners.	Addressed (scaffolded). New "Tool-specific resources" card placed at the top of the Resources page with three subsections: "How this tool works" (equation chain, MC method, sensitivity), "How to use the tool" (template prompts including the uncertainty_pct=0 convention you mentioned), "Preparing uncertainty inputs" (Approach 1 spreadsheet, distribution choice, time-series correlations). Marked as drafts for beta-testing — content can be expanded with your input.

"1 Data input" page

Comment	How it was addressed
22a. "Parameters" tab includes several parameters for manure management that would be more appropriate in the Manure_management tab because they are specific to manure management systems rather than animal sub-categories, e.g. EF3, EF4, EF5, Fracgas, Fracleach.	Mostly already done; clarification added. Looking at the current template: EF3 is already a per-MMS column on the Manure_Management sheet (with its own lower/upper/distribution columns), and so are Frac_GasMS_pct and Frac_LeachMS_pct. EF4 and EF5 are global IPCC scalars with no per-MMS variation — they have to stay as broadcast values, which is what the Parameters sheet is for. The Parameters-tab rows for `Frac_GASMS` and `Frac_LEACH_H` are kept as broadcast fallbacks (used only when the Manure_Management sheet doesn't supply per-MMS values). `EF3_PRP`, `Frac_GASM_PRP`, `Frac_LEACH_PRP` are pasture-deposition parameters (not MMS-storage) and stay on the Parameters tab.
22b. Some parameters are repeated in the "parameters" and "Manure_management" tab, e.g. Bo, and I'm not sure which tab the tool picks the value from. If it's specific to an animal type (e.g. Bo values vary between dairy and non-dairy) then may be better in the "parameters" tab.	Addressed. Bo removed from the Manure_Management sheet (the column was never consumed by the app — Bo always came from the Parameters sheet, which is correct because Bo varies by animal type, not by MMS). R/utils_template.R MM_COLS / MM_WIDTHS shortened by 1, downstream column-style indices shifted, hints row updated.
22c. Columns L and M are listed as "auto-fill" but did not auto-fill unless I placed the cursor at the end of the formula. If it does not auto-fill, this does not impact on the upload into the tool, but the autofill of lower and upper CIs is useful for the user to have some QC check when filling the templates before upload.	Addressed. Template generation now pre-computes lower/upper bounds as numeric cells when the row has known value + uncertainty_pct (i.e. for example rows and IPCC-default rows). Blank-template rows still use the Excel formula (since the user will fill them in). R/utils_template.R generate_template_openxlsx loop around the lower/upper write. No more openxlsx auto-recalc quirk on opening.
22d. In the "manure_management" tab, Bo has a value but no columns to fill uncertainty_pct or lower & upper CIs.	Addressed (by removal — see #22b). Bo no longer appears in Manure_Management, so the missing-uncertainty-columns issue is moot. Bo on the Parameters sheet already has its full uncertainty columns.
22e. "Parameter time series": I could not work out how to enter my data in the template. There are no columns for cattle type, aggregation level and sub-category. The filled example template looks like an aggregation or example for one production system only, so hard to reproduce from a more complex input dataset.	Addressed. Parameter_TimeSeries sheet now has `cattle_type`, `aggregation_level`, `sub_category` columns prepended before `year`. Example block filled with the dairy / Country X labels so users see the expected layout. Column widths / styles updated. Note: in this round the auto-correlation routine still treats all data rows as a single series; full per-group correlation computation can be added once you've validated this layout works for your data.

Comment	How it was addressed
23. Adjusting Cfi for winter temperature. You have the mean daily temperature in winter as an advanced option in '5. Simulate and results'. I think it would make sense to include it in the basic upload template and all processes thereafter; if a country is not using this adjustment, then it can be left blank and the app should give a default auto-fill value that negates the winter-specific adjustment of Equation 10.2, but including this option from the start would help make the tool IPCC-compliant. Also, some countries may apply this adjustment to some production systems (e.g. 'high Andes') but not others (e.g. lowland) in the same country.	Fully addressed (2026-05-18 follow-up). `Tw` is now sourced exclusively from the Parameters template as a per-sub-category column (IPCC Eq 10.2, default 20 °C = no adjustment). Wiring change: `run_mc_simulation` reads Tw from `samples` via `get_param("Tw", 20)` and passes it as a per-iteration vector through `ghg_emissions_vec`, which now broadcasts Tw the same way it does MilkPR / PRP fractions. Each MC iteration uses the matching row's Tw value, so the cold-climate Cfi adjustment can differ between e.g. a high-Andes and a lowland sub-category in the same inventory. The old global UI input on the Simulate panel was removed (it was a duplicate that also silently overrode the template value — wrong for IPCC compliance). The "Advanced options" panel now points the user at the Parameters sheet for Tw. The `pct_calving` slider stays in the UI as a single inventory-wide value. R/mc_simulation.R, R/calc_ghg_master.R, [R/app_ui.R](R/app_ui.R), R/app_server.R.

"2 QAQC" page

Comment	How it was addressed
24. General feedback: the tests were useful and picked up errors and omissions and gave useful advice on more appropriate PDFs.	Noted, no change required. Thank you.
25. Implementation of the benchmark deviation tests works – it raises lots of flags, but it often looks like the benchmarks used are not appropriate, possibly because IPCC sub-categories are defined differently from country-specific ones, or because the default values for the benchmark is from a different IPCC version (2006 or 2019) etc, e.g. benchmark references to Monni et al (2007) seem inappropriate unless they are also reflected in the IPCC guidelines. Overall, this test is useful, but perhaps just check how it applies default benchmark values to country-specific categories.	Addressed. Added a new `MONNI_BENCHMARK_PARAMS` set in R/utils_qaqc.R covering EF3_PRP, EF3_S, EF4, EF5, Frac_GASMS, Frac_LEACH_H and the two new PRP fractions. For these parameters, the `benchmark_deviation` check now reports `info` (not fail/warn) regardless of deviation, with an explanatory suffix ("benchmark is a Monni-2007/Penman-2000 mid-point; country-specific overrides are expected"). Added a new `info` severity level globally, with a blue ■ icon, a `n_info` count in `qaqc_summary`, and ranked between `warn` and `pass` in the sort order. Parameters with direct IPCC-table benchmarks (W, Milk, DE, Ym, Bo, …) still use the original fail/warn levels.

"3 Uncertainty" page

Comment	How it was addressed
26. The current set-up is to edit the flagged or failed items in the '3. Uncertainty' page. As a user, I found it logical to go back to the template, edit in the template and re-upload so that my final template contained a record of my final values. I also did this because there are more than 100 parameter values for just 1 production system in my pilot test-run, and it was easier to find and edit them in the template than using the search bar and editing in the tool. Maybe the current set-up is fine and the user guide can note both options for editing flagged/failed values…	Addressed. Added a bullet to the "How to use the tool" subsection on the Resources page documenting both workflows: (a) edit values in the in-app Parameters table (Tab 1) for quick one-off fixes; (b) edit the Excel template and re-upload to keep a saved record of your final values — better for runs with 100+ parameters. Both produce the same simulation inputs.

"4 correlations" page

I have no comments here, as I did not use the correlations option due to comment 22e above.

Comment	How it was addressed

"5 Simulate and results" page

Comment	How it was addressed
27. Emission sources to include: I suggest to list pasture deposition direct N2O and indirect N2O separately because they are separate IPCC reporting categories.	Addressed. Source selector on the Simulate tab now has two pasture keys: `pasture_n2o_direct` and `pasture_n2o_indirect`. `.apply_source_selection` in R/app_server.R and `filter_co2e` in R/trend_tab.R updated to honour each independently. Legacy `pasture_n2o` accepted as an alias so old saved selections still work. The skip-aggregation threshold updated from "<5 sources" to "<6 sources".
28. IPCC software aligned options: see comments 9 and 23 above.	See responses to #9 (deferred) and #23 (addressed via catalogue Tw row).
29. Possible bug? I checked the "Run uncertainty decomposition (AD/EF/Combined)" option, but there was no decomposition shown in my results. (I note it does work when running a simulation for the default in-app datasets, but didn't work for my uploaded custom dataset.)	Likely-cause fix applied. Root cause traced in R/mc_uncertainty.R `decompose_uncertainty`: when a custom upload has rows with `NA` in the `param_type` column (e.g. user left it blank), the line `ef_rows <- ad_specs$param_type == "coefficient"` produces a logical vector with NAs, and the subsequent `ad_specs$distribution[ef_rows] <- "constant"` throws "NAs not allowed in subscripted assignments" — silently aborting the AD-only run. The example datasets always have `param_type` populated, which explains why the bug only surfaced with custom uploads. Defensive fix: param_type is now coerced to character with `NA` → `"coefficient"` (and `parameter == "N"` → `"activity_data"`) before the subset. The Excel-download placeholder (#38) covers the residual case where decomposition still fails for some other reason — so the user gets a partial report instead of a broken download. Full reproduction with your `.xlsx` still useful to confirm the fix lands.
30. "Use 10,000 iterations for reliable results" – is there a basis for this statement? I seem to recall when I tested this with @Risk some years ago, 10,000 still led to quite large variation that was further reduced by more iterations. NB, in my test-run with Zimbabwe data, I ran the simulation twice with 25,000 iterations and got fairly consistent results.	Addressed. Wording softened on the Simulate tab to "10,000+ iterations recommended; check convergence by re-running with a different seed (1,000 is fine for quick testing)".
31. Minor edit: "What to do: Configure simulation settings on the left, then click 'Run Monte Carlo Simulation' at the bottom of the left-hand panel".	Addressed. Info panel on Simulate tab updated to include "at the bottom of the left-hand panel".
32. After running a simulation, the results are displayed at an inappropriate level of aggregation: results are given for each animal sub-category, whereas IPCC would require aggregation at the level of "dairy cattle" and "other cattle" for each emission source. Users may be interested in other levels of aggregation (e.g. comparing uncertainty for different production systems entered in the custom template "Parameters" tab column B) and potentially at the sub-category level (column C). Would it be possible before running the simulation to have the IPCC dairy vs other cattle as a default with the option for users to select additional lower levels of aggregation for results reporting?	Addressed. Added an "Aggregation level" `selectInput` on the Results tab with three levels — cattle_type (default, dairy / other), aggregation_level (production system), sub_category. New `.agg_results_by_level()` helper parses the system key (`cattle_type
33. First time, the simulations worked. Today I re-tried with the same uploaded template and got the error "--- Starting simulation --- Iterations: 5000 GWP: AR5 EF correlation: none (independent) ERROR: replacement has 0 rows, data has 5000 Custom data uploaded: 120 parameters loaded from uncertainty_template_ipcc2019.xlsx (country: Zimbabwe)" I could not determine the cause of the error, but hopefully the eventual user guide can document a number of error types and their solutions.	Defensive guard added. The error pattern "replacement has 0 rows, data has 5000" is the classic R `df[i] <- x_with_0_rows` failure. Most likely entry point: `run_inventory_simulation` in R/mc_simulation.R building the aggregator data frame via `rowSums(sapply(by_system, ...))` when one system's results happen to have zero rows (e.g. a Parameters row collapses to length 0 because `mean` is non-numeric). Hardened with two pre-flight checks: (a) `length(by_system) == 0` → stop with a descriptive error pointing the user at the Parameters sheet, and (b) any system whose results don't have `n_iter` rows → stop with a message naming the iteration count mismatch. Full repro with your `.xlsx` still useful to confirm this catches your specific case; if not, the error message will now name the exact mismatch instead of crashing in `data.frame()`.
34. Comment applicable to the initial display of results and all other reports subsequent: units (e.g. t CH4) are not always reported. Please make sure every column in the in-app result displays and the downloadable reports have units displayed.	Addressed. (1) By-System Breakdown columns renamed with explicit units: "Mean CH■ (t)", "Mean N■O (t)", "Mean CO■eq (t)", "MoE 95% (%)", "CV (%)", "CI lower (t CO■eq)", "CI upper (t CO■eq)". (2) By Reporting Category table same treatment. (3) CSV download (#36) now includes an explicit `unit` column ("t CH4" / "t N2O" / "t CO2eq") and an `emission_category` column in column A. (4) Word report Section 4 (#37, #39) now has a "Unit" column. (5) Headline value-boxes already showed " t" suffix.

"6 Sensitivity" page

Comment	How it was addressed
35. The tornado charts list parameters with the parameter abbreviations; to be really useful, it would present the results at the sub-category level of aggregation, e.g. cattle type_aggregation level_sub-category_parameter so that the user can for example identify that the most sensitive parameter is Ym for cows in the semi-intensive system, as opposed to Ym for any sub-category. Is it possible to revise either so that this is a default level of aggregation or to give the user the option to select the level of aggregation for the tornado charts?	**Partially addressed in this round.** The structural-bug part (Frac_LEACH_H / Frac_GASMS appearing in tornadoes for emission pathways where they are *mathematically absent* — your Section C "ISSUE TO HIGHLIGHT" findings) has been fixed. Root cause: when the selected output variable is structurally zero across all iterations (e.g. PRP direct N2O for an intensive-dairy run with `pct_pasture = 0`), the SRC/PRCC regression collapses on a constant outcome and returns numerical noise — surfacing whichever parameters happen to correlate with the zero vector. R/mc_sensitivity.R now checks `sd(output) > 1e-9` and, when the output is constant, returns an explanatory message instead of a tornado. The tornado renderer in R/app_server.R surfaces that message in place of the chart. This removes the cross-wiring artefact from the candidate-cause list for A1/A2. Still left to do: the broader request (per-cattle_type / per-sub-category aggregation in tornado labels) is queued for the follow-up round.

"7 IPCC reports" page and related downloaded report results

Comment	How it was addressed
36. The in-app reports show direct and indirect N2O from pasture deposit separately, which is good. The csv download and word reports do not. I suggest to always disaggregate these results if selected by the user.	Addressed. (1) CSV download now writes `pasture_n2o_direct` and `pasture_n2o_indirect` as separate rows (with their IPCC category labels in column A — "3.C.4 Pasture deposition - N2O direct" and "3.C.5 Pasture deposition - N2O indirect"). (2) Word Section 4 includes both as separate rows. (3) IPCC summary table now includes both pasture lines (was previously missing entirely).
37. The word report section 4 only reports by gas, not by emission category.	Addressed. `.results_flextable()` in R/utils_word_export.R rebuilt so Section 4 has one row per IPCC emission source (enteric CH■, manure CH■, MM N■O direct, MM N■O indirect, PRP N■O direct, PRP N■O indirect, totals) with an explicit "Unit" column and renamed numeric headers ("Emission category" / "Unit" / "Mean" / "CI lower" / "CI upper" / "CV (%)" / "MoE 95% (%)"). The helper falls back to `total_*` aggregator columns when per-system columns aren't present.
38. Excel download did not work.	Addressed. `export_results_xlsx()` in R/utils_export.R hardened against NULL or empty inputs. Previously crashed when `rv$ipcc_table` was NULL (which happens when AD/EF decomposition is skipped or fails for custom uploads), taking down the download handler with "site wasn't available". Now each sheet receives a `placeholder()` data frame with a "Note" cell explaining what's missing, so the workbook always writes. Also restructured the IPCC summary table to include pasture direct and indirect rows.
39. See Section C below for suggested revised template for word download.	Partially addressed. Section 4 of the Word report is rebuilt per your mock-up (per-source rows with units; pasture direct & indirect split). What's not yet implemented: the executive-summary "Activity data uncertainty / Emission factor uncertainty" tables you mocked up, and the per-cattle-type aggregation columns. Those will land in the next round alongside Andreas's confirmation on aggregation levels for partner rollout.

C) Overall feedback on a pilot test-run and suggestions on reporting

I input the values from Zimbabwe intensive dairy system (5 animal sub-categories) into the 2019 template and ran the simulation with the following settings:

- Single year
- No correlations
- 25000 iterations
- Random seed 42
- AR5, all emission sources selected

(I also ran it in @Risk with the same inventory values, and to the extent possible the same PDFs and uncertainty_pct inputs. Results for manure management will differ because I was not able to apply the Dirichlet constraint to MMS% values.)

My general experience was that using the tool was quite smooth and feasible to enter all the data required for 5 sub-categories. QAQC checks were useful (some of the benchmarks warnings seemed inappropriate but it was OK to ignore them and proceed). I did editing of 'fails' in the template and re-uploaded.

Comment	How it was addressed
C1. The 'headline' results in the coloured bubbles refer to total CH4 and total N2O, whereas an inventory user is always interested in IPCC-defined emission sources, e.g. enteric fermentation, manure management CH4 etc. The headline margin of error I guess refers to total CO2 equivalents, but again, this is not an IPCC-relevant number.	Addressed. The four headline value-boxes on the Results tab now show the IPCC source categories: "Enteric CH■ (t)", "Manure CH■ (t)", "Manure N■O (t)" (direct + indirect), "Pasture N■O (t)" (direct + indirect). Total CO■eq, 95% MoE, CV, Total CH■ and Total N■O moved to an inline footnote below the value-boxes (retained for sensitivity reference). New renderers `vb_enteric_ch4` / `vb_manure_ch4` / `vb_manure_n2o` / `vb_pasture_n2o` in R/app_server.R consume new inventory columns `total_enteric_ch4`, `total_manure_ch4`, `total_direct_n2o_mm`, `total_indirect_n2o_mm`, `total_direct_n2o_prp`, `total_indirect_n2o_prp` exposed by `run_inventory_simulation` in R/mc_simulation.R.
C2. The 'system breakdown' is per animal sub-category, whereas users initially require a breakdown by dairy/other cattle, and may be interested in second-level categories as defined in column B of the template for uploading. Results at cattle sub-category level would only be for the more advanced user.	Addressed. See #32 — Aggregation level dropdown defaults to `cattle_type`, drill-down available to `aggregation_level` or `sub_category`.
C3 Enteric sensitivity ranking matches @Risk (top 5 identical).	Noted as confirmation that the enteric pathway is correctly wired.
C4 MM CH4 tornado: MCF values rank high in @Risk but not in the tool.	**Partially addressed.** The zero-variance guard in R/mc_sensitivity.R removes the spurious-correlation source of artefacts but does not by itself explain the MCF ranking gap, which is more likely the Dirichlet-vs-fixed-MMS% difference noted under A1. Will be revisited once Zimbabwe data is re-run.
C5 MM Direct N2O: EF3 is influential in @Risk but not in the tool. ISSUE TO HIGHLIGHT: FRAC_LEACH_H is not a variable in the Direct N2O calculations.	**Cross-wiring artefact addressed in this round.** The "FRAC_LEACH_H in Direct N2O" finding was the spurious-rank pattern caused by SRC on a constant output. Fixed by the zero-variance guard described in #35. The EF3 ranking gap is the same issue as C4 (Dirichlet effect — to be confirmed with your data).
C6 MM Indirect N2O: Fracleach and Fracgas do not appear in the top variables in the tool, surprising given their large uncertainty ranges.	**Partially addressed.** Now that PRP and MM Frac parameters are separate (comment #10), the MM-side `Frac_GASMS` / `Frac_LEACH_H` are no longer being shared/masked by the PRP-side parameters in the sampler. Re-running your Zimbabwe pilot is expected to surface these in the MM-indirect tornado.
C7 Direct N2O pasture. ISSUE TO HIGHLIGHT: Fracgasms is not a parameter in the direct N2O from pasture equations.	**Addressed in this round.** Same fix as C5 — `Frac_GASMS` showing up as a top driver of direct PRP N2O was the same zero-output spurious-rank pattern (intensive dairy → `pct_pasture = 0` → direct PRP N2O is zero across all iterations). The tornado now shows an explanatory message instead of a numerically noisy ranking when the selected output is constant.
C8 Indirect PRP N2O. ISSUE TO HIGHLIGHT: FRAC_LEACH_H is not a variable in the Direct N2O calculations.	**Addressed in this round** (PRP indirect now uses the new `Frac_LEACH_PRP` from Table 11.3, not `Frac_LEACH_H` from Table 10.22). Re-run is required to confirm the tornado now shows `Frac_LEACH_PRP` (Table 11.3) for indirect PRP rather than the MM-side parameter.
C9 Excel download doesn't work.	Addressed. See #38 — `export_results_xlsx()` hardened against NULL inputs.
C10 CSV: put emission category in column A, separate pasture direct/indirect, allow user-defined aggregation.	Addressed. CSV download now has `emission_category` (column A), `unit` (column B), then the variable name and metrics. Pasture direct + indirect kept as separate rows with their IPCC reporting codes ("3.C.4" / "3.C.5"). User-defined aggregation: not exposed in the CSV itself — the CSV always lists the inventory-aggregate rows, but the Results tab tables now respect the cattle_type / aggregation_level / sub_category dropdown (see #32).
C11 Word report: rebuild per your proposed template with executive summary and per-source AD/EF tables.	Addressed. Two new flextables now appear immediately after the Executive summary in the Word report: "Activity-data uncertainty" and "Emission-factor uncertainty", each with one row per IPCC source category (enteric CH4 / manure CH4 / MM N2O direct / MM N2O indirect / PRP N2O direct / PRP N2O indirect) and the corresponding CV %. Pulled from the IPCC summary table via a new `.ad_ef_flextable()` helper in R/utils_word_export.R. Only rendered when AD/EF decomposition has been run (i.e. when `ipcc_table` is populated).
C12 Word report Section 7 — confirm the IPCC-reporting-context statement about column J.	Addressed (reworded). The "column J" reference removed from Section 7. The new wording describes the metric ("per-source combined-uncertainty CV %") and asks the user to cross-check the exact column letter against their national submission template, which avoids pinning a column that may differ between national versions. R/utils_word_export.R.

Summary of what changed in this round

Block A (calculation correctness):

- R/calc_manure_n2o.R — `calc_indirect_n2o_prp` now takes `Frac_GASM_PRP` / `Frac_LEACH_PRP` (IPCC 2019 Table 11.3) instead of reusing MM-side fractions (#10).
- R/calc_ghg_master.R — `ghg_emissions` / `ghg_emissions_vec` thread the new PRP-side fractions through.
- R/mc_simulation.R — `run_mc_simulation` reads `Frac_GASM_PRP` / `Frac_LEACH_PRP` from the sampled parameter set; `run_inventory_simulation` now exposes split MM/PRP totals per gas (`total_direct_n2o_mm`, `total_indirect_n2o_mm`, `total_direct_n2o_prp`, `total_indirect_n2o_prp`) alongside the existing combined totals.
- R/mc_sensitivity.R — zero-variance output guard. When the selected output is constant across iterations (e.g. PRP N■O when `pct_pasture = 0`), the function now returns an explanatory message instead of spurious SRC/PRCC ranks (resolves the "FRAC_LEACH_H in direct PRP tornado" artefact in C5/C7/C8).
- R/utils_ipcc_defaults.R — added the PRP defaults to `IPCC_DEFAULTS`.

Block B (UI, catalogue, QAQC):

- R/utils_template.R — `PARAM_CATALOGUE` extended from 24 to 28 parameters (adds `Frac_GASM_PRP`, `Frac_LEACH_PRP`, `EF3_S`, `Tw`). Wording fixed for Milk (#8), Fat (#13), C growth (#11), Ym (#12), CP (#7), EF4 / EF5 (#14). `param_tier` values renamed "technical" → "advanced" (#5) with back-compat. Template generator updated to write the new rows.
- R/utils_qaqc.R — added `Frac_GASM_PRP` / `Frac_LEACH_PRP` to `FRACTION_PARAMS`; new `MONNI_BENCHMARK_PARAMS` set downgrades benchmark-deviation flags for EF3 / EF4 / EF5 / Frac_* from fail/warn to a new `info` severity (#25); new `info` icon and counter in `qaqc_summary`.
- [R/app_ui.R](R/app_ui.R) — Home text edits (#1, #2, #4), analysis-mode toggle moved to "1. Data Input" (#3), level wording "core / technical" → "core / advanced" (#5), Resources page rebuilt with "Tool-specific resources" + Learning resources + IPCC 2019 link + Monni moved to case studies (#16, #17, #18, #19, #21), Simulate panel wording softened on iterations (#30) and Run-button location (#31), source selector split pasture direct/indirect (#27), Unicode subscripts applied throughout (#2), 4 IPCC-source value-boxes replace Total CH■ / N■O (C1), Aggregation level selector added on Results tab (#32, C2).
- R/app_server.R — `.apply_source_selection` updated for 6-source selector (#27); 4 new IPCC-source value-box renderers (C1); `.agg_results_by_level()` helper aggregates by chosen level (#32); `results_by_system` and `results_by_category` tables consume the aggregator and add explicit units (#34); CSV download adds `emission_category` + `unit` columns (#36, C10); tornado renderer surfaces zero-variance message (#35 structural).
- R/trend_tab.R — `filter_co2e` updated to accept the split pasture keys.

Exports:

- R/utils_export.R — `export_results_xlsx()` hardened against NULL inputs so Excel download no longer crashes when decomposition was skipped (#38). `format_ipcc_table()` extended to include pasture direct + indirect rows (#36, #39).
- R/utils_word_export.R — `.results_flextable()` rebuilt with one row per IPCC emission source plus a "Unit" column (#37, #39, C11); `.pretty_var()` extended; new `.unit_for_var()` helper.

Code-base audit follow-up (2026-05-18, three parallel deep-dive reviews):

Triggered after the Dirichlet removal + N-excretion fix to look for any other non-IPCC patterns. Three parallel audits (calc engine / Monte Carlo / template+QAQC) surfaced both real issues and false positives. Real issues fixed:

- **Removed unused `DE` argument from `calc_n_excretion`.** Now that the DE factor is no longer applied to N intake (per the A1/A2 fix below), the parameter was dead weight in the function signature — could confuse future callers. Removed from both the function and its single call site in `ghg_emissions`. R/calc_manure_n2o.R, R/calc_ghg_master.R.
- **Defensive bounds checks** in `calc_n_excretion` (MilkPR ∉ [0,10]), `calc_volatile_solids` (ASH ∉ [0,1), UE ∉ [0,1]), and `calc_nem` (Tw < -50 °C). Each emits a `warning()` rather than crashing — preserves backwards-compat but surfaces data-entry errors. R/calc_manure_n2o.R, R/calc_manure_ch4.R,

R/calc_energy.R.

- **`nearPD` silent-adjustment warning.** Every time the tool builds a correlation matrix (user-supplied via correlations tab, IPCC preset, or expanded from a partial spec), `Matrix::nearPD` silently re-projects it onto the nearest positive-definite matrix. Previously this could change user-specified correlations by up to ±0.1 without notice. New `.repair_corr()` helper emits `warning()` when the adjustment exceeds 0.05. R/mc_sampling.R.
- **Whitespace + case normalisation on upload.** Trailing spaces in `parameter` column ("N " instead of "N") were silently dropping rows because the catalogue match failed. Case differences ("Dairy" vs "dairy") were splitting one cattle group into two systems_data entries. `parse_uploaded_template` now `trimws()` on parameter / cattle_type / aggregation_level / sub_category / mms_type and `tolower()` on cattle_type and distribution. R/utils_template.R.
- **Lognormal convention documented.** Empirically verified that the lognormal sampler uses `mean_val` as the **median** (`mu_log = log(mean_val)`), which puts the realised arithmetic mean ~10–25 % above `mean_val` for the IPCC asymmetric defaults (EF3_PRP, EF4, EF5, Frac_LEACH_H). This is consistent with the IPCC GPG 2000 / Penman 2000 / Monni 2007 convention where "central value" for a skewed distribution is the geometric mean / median, not the arithmetic mean. Added a clear comment block at the top of R/utils_distributions.R so future readers can defend the choice.

Audit false-positives (verified and dismissed): catalogue param_type length (28 columns × 28 elements confirmed); `Frac_GasMS` / `Frac_GASMS` "case mismatch" (these are deliberately different columns — Parameters has the broadcast scalar `Frac_GASMS`; Manure_Management has the per-MMS `Frac_GasMS_pct`); the beta-variance formula (the `(upper-lower)` cancellation is mathematically correct because the variance is computed on the [0,1]-normalised beta scale before being rescaled back).

A1 / A2 root cause — N excretion DE bug (2026-05-18): With your `uncertainty_template_ipcc2019_ZIM_v2.xlsx` we traced the 33 → 25 t direct-N2O gap to a real bug in `calc_n_excretion`: it applied the DE digestibility factor inside the N-intake step, contradicting IPCC 2006 Eq 10.32 / 2019 Refinement Eq 10.32A which use **gross** feed energy for N intake. DE belongs in the volatile-solids equation (Eq 10.24), not in N intake. The mix mis-weighted N intake by ~25-30 % and cascaded into direct/indirect N2O. Fixed in R/calc_manure_n2o.R. After the fix, tool output matches Excel to within ±10 % on every emission source (direct N2O MM 30.71 vs 33.23; indirect N2O MM 15.42 vs 16.89; manure CH4 603 vs 573; enteric CH4 2660.22 vs 2660.88 — exact). A secondary issue in your specific template (`DINT_heif` vs `DINT_heifer` between Parameters and Manure_Management sheets) caused the heifer group's MMS allocation to fall back to default — added an upload-time warning in R/app_server.R so this kind of mismatch surfaces in the simulation log.

Two further IPCC-alignment fixes from the calc-engine audit:

1. `MilkPR` **plumbing fix.** The catalogue parameter `MilkPR` (milk protein %, IPCC 2006 Table 10.11) was previously inert — `calc_n_excretion` in R/calc_manure_n2o.R hardcoded `milk_protein <- 3.3` regardless of what the user supplied or what the Monte Carlo sampled. Now `MilkPR` is a function argument with default 3.3, threaded through `ghg_emissions` and `ghg_emissions_vec` (R/calc_ghg_master.R), and pulled from samples in `run_mc_simulation` (R/mc_simulation.R). Regional / sub-category overrides and uncertainty sampling on milk protein now actually affect the N-excretion result. Default 3.3 preserves prior behaviour for templates that don't supply the parameter.

2. **AR6 GWP for CH■ corrected from 27.9 → 27.0.** The previous AR6 value didn't match any published IPCC AR6 figure. IPCC AR6 WG1 Table 7.15 publishes CH■-fossil = 29.8 and CH■-non-fossil = 27.0. Cattle CH■ (both enteric and manure) is biogenic / non-fossil, so 27.0 is the correct value. N■O = 273 unchanged; AR4 (25 / 298) and AR5 (28 / 265) unchanged. **Impact:** runs that selected AR6 will have CO■eq totals shift by approximately ((27.0–27.9)/27.9) ≈ –3.2 % of the CH■ contribution. AR5 is the default, so the headline-default behaviour is unaffected. UI label on the GWP dropdown updated accordingly. R/utils_ipcc_defaults.R and [R/app_ui.R](R/app_ui.R).

Methodology change — Dirichlet MMS-allocation sampling removed: the previous code path (introduced in Round 7 T4.21) sampled per-iteration MMS% allocations from a Dirichlet on the simplex, controlled by a "MMS allocation precision (Dirichlet concentration)" field on the Simulate tab. Per discussion: this is not explicitly cited in IPCC 2006 / 2019 Refinement guidance, and the A1/A2 diagnosis already proved Dirichlet had no effect on the mean of any emission source (theorem: Dirichlet preserves marginal means when per-MMS coefficients are constant). MMS% is now deterministic across iterations, matching the IPCC Inventory Software's behaviour. Files touched: R/mc_sampling.R (removed `sample_dirichlet_simplex`); R/calc_ghg_master.R, R/mc_simulation.R (removed `mms_fractions_matrix` parameter and the use_dir branch); R/app_server.R (removed the Dirichlet matrix build); [R/app_ui.R](R/app_ui.R) (removed the `mms_concentration` numericInput); diagnose_zimbabwe_n2o.R (run "C" removed, header updated to explain the historical context). The diagnosis script's run B re-validates the simulation engine returns identical numbers to before the removal.

Final round added:

- R/utils_template.R — added `BW → W` alias (#6 partial); reworded Milk definition to spell out per-lactating-animal convention (#9 partial); removed unused Bo column from Manure_Management (#22b, #22d); pre-computed numeric lower/upper bounds for example template rows so they no longer rely on Excel formula auto-recalc (#22c); added `cattle_type` / `aggregation_level` / `sub_category` columns to Parameter_TimeSeries (#22e).
- R/mc_uncertainty.R — defensive coercion of `param_type` (NA → "coefficient") at the top of `decompose_uncertainty` (#29).
- R/mc_simulation.R — pre-flight checks in `run_inventory_simulation` against empty `by_system` and per-system row-count mismatch (#33).
- R/utils_word_export.R — new `.ad_ef_flextable()` helper; AD- and EF-uncertainty per-source tables added immediately after the Word Executive summary (C11); Section 7 "column J" reference removed (C12).
- [R/app_ui.R](R/app_ui.R) — workflow note added to Resources page documenting both edit paths (#26).

What is still blocked and needs your input:

- Your populated `uncertainty_template_ipcc2019.xlsx` so we can finish the A1/A2 calc-correctness investigation and confirm the #29 / #33 hardening catches your specific failure mode.
- Confirmation whether your @Risk run included AD uncertainty or EF-only (A1).
- Confirmation whether your Zimbabwe `Milk` input is per-lactating-animal or sub-category-average (#9 / A1 hypothesis 3).
- Sign-off on the W → BW full rename (#6) before we flip the canonical name across the catalogue, UI, and example templates. The `BW` alias is already in place so users can upload with that name.
- Any case-study national-inventory links you wanted to compile (#20).

Items intentionally not revisited in this round (Block A partial work already done — see earlier rows): A1, A2, #10, the structural-bug part of #35, C4–C8.